LEVIATHAN QUANT SYSTEM OPERATOR FEEDBACK CONSOLIDATED REPORT

Engine 2 (Macro Shape Chain) + Engine 3 (Trade Setup Architecture)

CRITICAL ENGINEERING REFERENCE — Operator Corrections & Specifications

Date: 2026-05-26

Classification: INTERNAL — Engineering Handover

System	Leviathan Ultra-Quant / Trident Trading Engine v5.0.0
Document Purpose	Consolidate ALL operator feedback on trade setups and macro shape chains for engineer handover
Engine 2 Status	ACTIVE — Macro Shape Chain semi-accurate; operator corrections needed for full accuracy
Engine 3 Status	DISABLED — Graded D in postmortem; complete rebuild specified but not yet implemented
Source Material	April 20-24 Postmortem, VLM screenshot reviews, operator live corrections during report generation

1. Executive Summary

This document consolidates every piece of feedback the operator provided on how to correctly construct trade setups (Engine 3) and how to correctly interpret and forward-map macro shape chains (Engine 2). The feedback was given during the April 20-24 postmortem review, through VLM screenshot comparisons of generated reports versus actual market outcomes, and through live corrections during subsequent report generation sessions. This document is the authoritative reference for the next engineer — every correction, every principle, and every "do NOT do this" rule documented here came directly from the operator's trading expertise and real-market validation.

The key finding is that Engine 3 (Trade Setup Architecture) was graded D because it produced a single wrong-direction trade setup for an entire five-day week that contained at least three high-probability setups. The root causes were: (1) no counter-trend validation at hard-wall zones, (2) single-setup architecture instead of multi-setup, (3) vague entry/SL/TP anchoring instead of precise key-level anchoring, and (4) regime-incorrect TP methodology (fib wormhole in SS regime instead of zone-edge targeting). These were not edge cases — they were fundamental architectural failures that the operator identified through systematic comparison of system predictions against actual trading outcomes.

Engine 2 (Macro Shape Chain) was graded more favorably (A for IP zones, A-/B+ for shape chain) but had specific accuracy issues: the May 18-22 chain prediction was wrong (BS→R→PBS predicted, actual was BS→RWL→SS→WLS) because a critical IP zone at 99.43-99.67 was not properly mapped. The operator taught that macro shape chains are deterministic when anchored to IP zones — the shapes are not subjective pattern labels but mechanically defined by IP zone interactions.

2. Engine 3 — Trade Setup Corrections (Operator Teaching)

This section documents every correction the operator made to Engine 3 (Trade Setup Architecture). Engine 3 was graded D in the April 20-24 postmortem because it produced a single wrong-direction trade setup for an entire week that contained at least three high-probability setups. The operator's corrections address four fundamental root cause failures and establish ten non-negotiable rules for the rebuilt architecture.

2.1 The April 20-24 Trade Setup Failure — Complete Analysis

The system produced ONE trade setup for the week of April 20-24: a SHORT at the Equilibrium Zone (98.29) with SL at 98.50 and TP at 97.70 (2.8x RRR). This was directionally wrong — the entire week was a BUY trend. The system's 5-Check Direction Validation returned 5/5 SHORT consensus, but the actual market went LONG from the exact zones the system identified.

System vs Actual Comparison:

Parameter	System Prediction	Actual Outcome
Direction	SHORT	BUY trend all week — SHORT stopped out
Entry	98.29 (Equilibrium Zone edge)	98.44 (key level at 98.365 inside the zone was the real anchor)
Stop Loss	98.50 (above zone + 0.09 buffer)	98.525 (1 pip above next key level at 98.44, not arbitrary zone buffer)
Take Profit	97.70 (Fib wormhole target)	98.02 (LSZ edge at Mid-Range Support) or 98.00 (zone edge)
RRR	2.8x	4.06x (using correct entry/SL/TP)
Outcome	Loss (SL hit)	The SHORT triggered from the predicted zone but was overextended and poorly anchored

2.2 Three Missed Setups the Operator Identified

The operator identified that the week contained at least three clean, high-RRR setups that the system completely missed:

#	Direction	Entry	Stop Loss	Take Profit	RRR	Day	Notes
1	SHORT	98.44	98.525	98.02	4.06x	Mon-Tue	Entry at key 98.365 level inside Equilibrium zone. TP at LSZ edge of Mid-Range Support. The SHORT the system predicted DID happen, but from wrong entry with wrong TP.
2	LONG	98.02	97.87	98.77	5.0x	Tue-Thu	Entry at LSZ edge (98.02) of Mid-Range Support. SL 1 pip inside next LSZ. TP at edge of Upper Resistance zone. Clean 2-day swing scalp.
3	LONG	98.29	98.185	98.77	4.57x	Wed-Thu	Reinforcement LONG from the same Equilibrium Zone the system had as SHORT entry. Entry at 98.29, SL at 98.185 (1 pip below zone), TP at 98.77 (resistance edge).

2.3 Four Root Cause Failures — Operator Diagnosis

FAILURE 1: Direction Bias — SHORT Consensus vs BUY Reality

The 5-Check Direction Validation protocol returned 5/5 SHORT consensus. Every single check (4H trend, BoM alignment, LSZ direction, zone failure, macro shape) pointed to SHORT. The actual week was a sustained BUY trend. The root cause is that the direction validation protocol is backward-looking by design. It looks at what HAS happened (prior 4H trend was bearish, prior daily closes were declining) and projects the same trend will continue. It does not account for trend exhaustion signals, regime transition timing, or mean-reversion probabilities at extreme support levels.

OPERATOR CORRECTION: When DXY is at 98.10, sitting just 13 pips above a Key Support Band with ZFP 12% (a Hard Wall), the probability of a bounce is significantly higher than the probability of a continuation through that wall. The direction validation system needs a COUNTER-TREND CHECK that evaluates mean-reversion probability when price is near a high-confluence support/resistance zone. This is now codified as PostMortem Fix P1.

FAILURE 2: Single Trade Architecture — Only 1 Setup for 5 Days

The most glaring failure is that Engine 3 only produced ONE trade setup for an entire 5-day trading week. The system prompt instructed Engine 3 to produce "primary" and "secondary" trade setups, which it did (SHORT primary, LONG secondary). However, the LONG secondary was treated as a contingency trade ("if price reaches Key Support, then LONG") rather than an independent, co-equal trade opportunity. In reality, BOTH the SHORT from Equilibrium AND the LONG from Mid-Range Support were valid simultaneous setups, not sequential contingencies. Furthermore, the system completely failed to identify the Wednesday LONG reinforcement from the Equilibrium Zone, which was a third independent setup.

OPERATOR CORRECTION: The root cause is architectural — Engine 3 operates on a "primary chain" model (SS then BS then R then WL) where each trade is assumed to follow the previous one. In a Swing Scalps regime, MULTIPLE independent setups exist simultaneously at different zones, and the system should generate ALL of them, not just the one that fits the primary chain narrative. This requires a MULTI-SETUP ARCHITECTURE where every IP zone with ZFP < 30% produces both a zone-rejection trade AND a zone-breakdown trade as independent setups. This is now codified as PostMortem Fix P2.

FAILURE 3: No Key Levels Inside/Outside Zones for Entry Anchoring

The system set the SHORT entry at 98.29 (the Equilibrium Zone edge) as a zone-edge entry. However, the actual precise entry was at 98.44, which is a key level inside the zone that the system did not identify or map. Similarly, the stop loss was set at 98.50 using an arbitrary "zone + buffer" formula, but the correct SL was at 98.525, anchored to the next key level above.

OPERATOR CORRECTION: The system lacks an internal key-level mapping within each IP zone. When a zone spans 98.29-98.41 (0.12 DXY wide), there are internal structure points (like 98.365, 98.38, 98.44) where price has previously reacted. These micro-levels must be identified and used for precise entry/SL anchoring, rather than using the zone edge as a blunt instrument. This requires adding a Step 0.6 "Intra-Zone Key Level Scan" to the IP Zone methodology, which would identify all minor pivot points, prior reaction highs/lows, and option strike conversions within each IP zone boundary.

FAILURE 4: Take Profit Overestimation — Fib Wormhole vs Zone Edge

The system set the take profit at 97.70 using a Fibonacci wormhole calculation (0.618 retrace to 1.618 extension from the swing high 98.41 to swing low 97.63). This target was 0.40 DXY below the entry, requiring price to break through the Mid-Range Support zone (98.00-98.12) and the LSZ at 97.97-98.02 to reach it. In a Swing Scalps regime where both bounding zones have ZFP less than 30%, the maximum expected move is to the edge of the nearest opposing zone, NOT through it.

OPERATOR CORRECTION: The correct TP was at 98.02 (LSZ edge of Mid-Range Support) or 98.00 (zone edge), which offered a clean 4.06x RRR with the correct entry/SL. The Fibonacci wormhole is a valid targeting tool for Breakout Scalps and Trend Continuation scenarios, but it should NOT be the default TP method for Swing Scalps. In SS regime, the default TP should be the edge of the nearest opposing IP zone or LSZ, with Fib wormhole as a secondary "extended target" only if the zone is breached. This REGIME-CONDITIONAL TP LOGIC must be added to Engine 3.

2.4 Operator Rules for Correct Trade Setup Construction

The following rules were taught by the operator through the postmortem and VLM screenshot review process. These are NON-NEGOTIABLE rules that must be followed in Engine 3's rebuilt architecture:

TS-1	Entry must be at a SPECIFIC key level, NOT a zone edge. "Entry: 98.44 (key level 98.365 inside Equilibrium Zone)" is correct. "Entry: 98.29 (Equilibrium Zone edge)" is wrong. — Origin: Failure 3, operator screenshot comparison
TS-2	Stop Loss must be anchored to the NEXT key level beyond the zone, NOT an arbitrary buffer. "SL: 98.525 (1 pip above key level 98.44)" is correct. "SL: 98.50 (zone + 0.09 buffer)" is wrong. — Origin: Failure 3, operator screenshot comparison
TS-3	Take Profit in SS regime MUST default to the nearest opposing IP Zone or LSZ edge. Fib wormhole is ONLY acceptable as an extended target AFTER the primary zone-edge TP. "TP: 98.02 (LSZ edge, Mid-Range Support)" is correct. "TP: 97.70 (Fib wormhole)" is wrong for SS regime. — Origin: Failure 4, operator live correction
TS-4	When price is within 0.20 DXY of a HEAVY zone (ZFP < 15%), a COUNTER-TREND CHECK must evaluate mean-reversion probability. The direction validation CAN be overridden by counter-trend probability at hard walls. — Origin: Failure 1, operator postmortem analysis
TS-5	Generate ALL valid trade setups across ALL zones SIMULTANEOUSLY, not just the primary chain. Each IP zone with ZFP < 30% should produce BOTH a zone-rejection trade AND a zone-breakdown trade as INDEPENDENT setups. — Origin: Failure 2, operator postmortem analysis
TS-6	In a Swing Scalps regime, BOTH a SHORT from the ceiling zone AND a LONG from the floor zone are valid CO-EXISTING setups, not sequential contingencies. — Origin: Failure 2, operator teaching
TS-7	The Intra-Zone Key Level Scan (Step 0.6) is MANDATORY. It identifies all minor pivots, prior reaction levels, and option strike conversions within each IP zone for precise entry anchoring. — Origin: Failure 3, operator specification
TS-8	SL placement: for longs, SL = demand-side LSZ bottom edge minus 3 pips buffer. For shorts, SL = supply-side LSZ top edge plus 3 pips buffer. If no LSZ exists within 30 pips, REJECT — no valid SL anchor. — Origin: Engine 3 rebuild spec (R3.3)
TS-9	TP placement: TP1 = nearest opposing IP Zone boundary. TP2 = second opposing boundary (if within 80 pips). RRR = (TP1 - Entry) / (Entry - SL). Minimum 2.0:1 for trend, 2.5:1 for counter-trend. — Origin: Engine 3 rebuild spec (R3.4, R3.5)
TS-10	Maximum 3 setups per report. Quality over quantity. If zero setups meet all criteria, output "No qualifying setups this week." — Origin: Engine 3 rebuild spec (R3.10)

2.5 Engine 3 Rebuild Architecture (6 Phases)

The operator approved the following 6-phase architecture for Engine 3's rebuild. This must be implemented before Engine 3 is re-enabled:

Phase 1: ZONE SELECTION

Input from Engine 1 IP Zone Map. For each zone with Pressure >= MODERATE (7/14+), classify as potential ENTRY zone or TP target. Entry zones = zones where price is approaching (within 15 pips). TP zones = opposing zones in trade direction.

Phase 2: SHAPE ALIGNMENT

Input from Engine 2 Macro Shape Chain. Check if predicted shape supports trade at this zone. BS = directional supported. WL = counter-trend fade supported, directional DOWNGRADED. SS = no directional trade, scalp only. R = counter-trend entry supported. DISAGREEMENT LOG: If Engine 2 contradicts trade direction, note disagreement and reduce confidence by 20 points.

Phase 3: CONFLUENCE SCORING

Count 9-Pillar pillars firing at entry level. Minimum 6/14 required. Cross-pillar bonuses: Option + Structure overlap +1, 3+ pillars in same 20-pip band +1, Repeated Magnet at entry +1. Max score: 14. Score < 6 = REJECT.

Phase 4: SL/TP ANCHORING

SL anchored to nearest LSZ edge on wrong side. TP1 anchored to nearest opposing IP Zone boundary. TP2 to second boundary if within 80 pips. RRR < 2.0:1 = REJECT. Counter-trend RRR < 2.5:1 = REJECT.

Phase 5: CONFIDENCE SCORING

Base = (Confluence/14)*40 max 40. Option bonus = (Notional/3.5B)*20 max 20. Shape alignment = 20 if aligned, 0 if disagreeing. Red folder clearance = 10 if clear. ZFP bonus = (1-ZFP)*10 max 10. Total 0-100. Score < 65 = REJECT. Red Folder within 30 min = DEFER.

Phase 6: OUTPUT FORMATTING

Each setup as KV table: Direction, Entry (exact level + source), SL (exact level + LSZ anchor), TP1/TP2 (exact level + zone anchor), RRR, Confidence score, Confluence count/pillars, Shape alignment, Red Folder status, Anchor zone, Risk note.

2.6 Engine 3 Failure Modes to Prevent

Table of 6 failure modes that caused the D grade:

Failure Mode	Description	Impact	Prevention
Garbage In, Garbage Out	Engine 3 consumed unfiltered option data. Sub-threshold strikes used as entry/exit anchors.	FATAL — entries at levels with no magnet effect	Filter Checkpoint (D1) before Engine 3 receives any data
No Confluence Floor	Setup generated if ANY single pillar suggested entry. 2/14 treated same as 10/14.	CRITICAL — low-confluence = noise	Minimum 6/14 floor (R3.2)
Vague SL/TP	SL at "nearby support" without LSZ anchor. TP at arbitrary numbers.	CRITICAL — wrong SL = stopped out for no reason	LSZ-anchored SL (R3.3), IP Zone-anchored TP (R3.4)
No RRR Gate	Setups with 0.8:1 RRR presented alongside 3.5:1 without filter.	HIGH — low-RRR wastes capital	Minimum 2.0:1 trend, 2.5:1 counter-trend (R3.5)
No Temporal Alignment	Bullish setup generated when Engine 2 predicted whiplash.	HIGH — right level, wrong time	Shape alignment check (R3.6)
No Red Folder Deferral	Setups generated for times with high-impact data releases.	MODERATE — trading into NFP is gambling	30-minute deferral window (R3.7)

3. Engine 2 — Macro Shape Chain Corrections (Operator Teaching)

This section documents every correction the operator made to Engine 2 (Macro Shape Chain Map). Engine 2 performed better than Engine 3 but had specific accuracy issues in chain sequencing, timing, and IP zone anchoring. The operator's teaching reveals that macro shapes are NOT subjective pattern labels — they are mechanically defined by IP zone interactions, and the classification becomes deterministic once IP zone metrics are known.

3.1 Engine 2 Performance Assessment

Engine 2 (Macro Shape Chain Map) performed better than Engine 3 but had specific accuracy issues. The operator's assessment:

IP Zone Mapping	A/A+ (4 of 5 zones precise, 1 too wide)
LSZ Mapping	A+/A (5 of 6 zones accurate, 1 bottom edge off)
Macro Shape Chain	B+/A- (shape identification mostly correct, chain sequencing had errors)
Critical Failure	The May 18-22 chain prediction was WRONG — this was the critical teaching moment

3.2 The May 18-22 Case Study — Chain Prediction Was Wrong

Without IP zones mapped, the May 18-22 price action appeared to follow the chain BS→R→PBS. The Breakout Surge was followed by a brief Recovery, and it appeared the surge would continue piercing higher. However, this analysis was WRONG because the IP zone at 99.43-99.67 was not properly mapped.

The 99.43-99.67 zone was a Heavy to Extreme pressure resistance zone with high structural confidence (likely a Confluent Ceiling or Repeated High Magnet from Engine 1 Pillar 5 Step 0.5). This zone had ZFP well below 30%, meaning it was highly likely to HOLD against a breakout attempt. The "Recovery" that appeared to be forming was actually price approaching this strong IP resistance zone, and the "PBS" that appeared to be starting was actually the zone rejecting the surge.

CORRECT CHAIN WITH IP ZONES: BS→Reverse Whiplash→SS→Whiplash Storm. The BS reached the 99.43-99.67 zone, was violently rejected (Reverse Whiplash), then price consolidated in a volatile range (SS transitioning to WLS) as the market oscillated between the lower IP zones and the rejected ceiling. Had the IP zone been mapped beforehand, the incorrect BS→R→PBS prediction would never have been made, because the ZFP of the 99.43-99.67 zone would have flagged that a breakout was unlikely there.

OPERATOR TEACHING: This case study demonstrates the core principle — macro shapes are NOT subjective pattern labels. They are mechanically defined by IP zone interactions. The classification is deterministic once the IP zone metrics are known. The shape taxonomy (SS, BS, R, WL, WLS) is defined by how price interacts with specific zone types, not by visual pattern recognition alone.

3.3 The Four Base Macro Shapes — Operator-Verified Definitions

Swing Scalps (SS)

Definition: Price oscillates within a range bounded by IP zones above (resistance ceiling) and below (support floor). Both ceiling and floor zones have ZFP below 30%. Option cage effect creates repeated Spring Release bounces.

OPERATOR CORRECTION: SS lasting more than one full trading session without a breakout attempt is classified as SS. If a recovery phase persists for more than one full session, it is reclassified as SS rather than Recovery. This temporal distinction is CRITICAL — Recovery is transient (half-session to one session), while SS is sustained (multi-session consolidation).

Subtype — Double Whiplash Pattern: Within SS, two rapid whiplash moves in succession within the range. Labeled "SS (Double Whiplash)" indicating heightened volatility, often preceding a breakout attempt.

Breakout Surge (BS)

Definition: Price breaks decisively through an IP zone with ZFP above 45%, confirmed by sustained move beyond the zone (15+ minutes). The target is anchored to the nearest Structural Extreme Magnet via Measured Move Projection.

OPERATOR EMPHASIS: The 15-minute confirmation is NOT optional. A wick or brief spike through a zone is NOT a breakout. The Measured Move target is always anchored to the nearest Structural Extreme Magnet, NOT a blind projection.

Variant — Piercing Breakout Surge (PBS): So powerful it pierces directly through one or more intermediate IP zones without pausing. Occurs when failing zone had ZFP above 70% and macro catalyst was exceptionally strong. PBS almost always produces a Reverse Whiplash or Whiplash Storm at its termination point.

Recovery (R)

Definition: Temporary stabilization or mild retracement after BS or WL. Price consolidates at a mid-range IP zone (Moderate to Moderate-Heavy pressure, ZFP 20-40%).

OPERATOR CORRECTION: Duration is half-session to one full session. If stabilization persists for MORE than one full session, it is reclassified as SS. The transition from R to SS represents a structural upgrade where the market has accepted the new range. This is a key timing distinction that the operator emphasized repeatedly.

Variant — Recovery Surge (RS): Instead of slow consolidation, Recovery takes the form of a sharp directional move back toward the pre-breakout level. RS signals the prior BS may have been a false breakout, increasing the probability of a Reverse Whiplash next.

Whiplash (WL)

Definition: Rapid violent directional move that immediately reverses at an IP zone, creating a spike-and-reject pattern. Occurs at Hard Walls or Institutional Magnets with ZFP below 20%. The key characteristic is the REJECTION at the IP zone, not the initial surge.

OPERATOR EMPHASIS: Whiplash is guaranteed at Hard Walls with ZFP below 15%. The Liquidity Sweep Zone designation from Engine 1 is the primary identifier for Whiplash-producing zones. The market surges toward the zone to trigger stop-loss orders clustered at the prior extreme, then reverses as institutional limit orders defend the level.

Variants:

Reverse Whiplash (RWL): Whiplash in the opposite direction of the preceding BS. After upside BS, RWL is violent downside rejection. Most common follow-on shape after BS, forming the BS→RWL chain. When RWL rejects back through the BS origin, it signals false breakout.

Piercing Whiplash (PWL): Initial surge so powerful it pierces through multiple IP zones before rejection. Rejection point is typically a Deep Hard Wall or Institutional Magnet.

Piercing Reverse Whiplash (PRWL): RWL with piercing characteristics. Strongest reversal signal in the taxonomy. Almost always leads to new SS phase or Whiplash Storm.

Whiplash Storm (WLS)

Definition: Extended sequence of rapid explosive Whiplash moves across multiple sessions. On 4H timeframe, WLS appears as SS with dramatically increased volatility. On every timeframe below 4H, violent successive whiplashes in both directions.

OPERATOR TEACHING: WLS almost always resolves into a Breakout Surge when one of the bounding IP zones finally fails. The transition WLS→BS is one of the HIGHEST-CONFIDENCE forward-mapping predictions because the prolonged oscillation builds institutional positioning pressure that must eventually release directionally. This is a key trading signal.

3.4 Shape Classification Decision Matrix (Operator-Verified)

Priority-ordered classification matrix:

Priority	Condition (IP Zone Metrics)	Shape	Confidence
1	Price breaks through zone with ZFP > 45% AND confirms 15+ min; Measured Move target exists	BS (or PBS if pierces multiple zones)	HIGH
2	Price surges to zone with ZFP < 20%, briefly pierces, then rejects; Liquidity Sweep Target identified	WL (or PWL if multiple zones pierced)	HIGH
3	After BS/WL: price stabilizes at mid-range zone (ZFP 20-40%); duration < 1 session	R (or RS if sharp retracement)	MODERATE
4	3+ IP zones in narrow range (0.30-0.50 DXY); price oscillating violently; 4H shows consolidation	WLS (or Minor WLS if 1-2 sessions)	HIGH
5	Price oscillating between Heavy/Extreme ceiling and floor zones; both ZFP < 30%; duration > 1 session	SS	HIGH
6	After BS: violent reversal in opposite direction through prior origin; Hard Wall rejection at origin	RWL (or PRWL if piercing)	MODERATE-HIGH

OPERATOR EMPHASIS: This matrix must be applied IN SEQUENCE — check highest-priority condition first. The classification is NOT subjective — each shape is determined by quantifiable IP zone metrics produced by Engine 1.

3.5 Shape Transition Rules (Operator-Verified)

Table of transition rules:

Current Shape	Next Shape	IP Zone Trigger	Probability
SS	BS	A bounding zone with ZFP > 45% fails with 15+ min confirmation	HIGH (when ZFP > 60%)
SS	WLS	3+ zones in narrow range; volatility triggers amplify oscillations	MODERATE
BS	R	Price reaches Measured Move target zone (ZFP 20-40%); momentum fades	HIGH
BS	RWL	Price reaches Hard Wall / Institutional Magnet; strong rejection	HIGH (at Hard Walls)
BS	PBS	Zone fails with extreme momentum; target is distant Structural Extreme	MODERATE (at high ZFP)
R	SS	Stabilization persists > 1 session; both bounding zones hold	MODERATE-HIGH
R	WL/RWL	Recovery reaches opposing IP zone; Whiplash rejection	MODERATE
R	BS (2nd leg)	Recovery at mid-range zone that also fails; cascade continues	MODERATE
WL	RWL	Whiplash rejected at zone; opposite direction momentum builds	MODERATE-HIGH
WL	SS	Whiplash exhausts momentum; price consolidates near rejection point	MODERATE
RWL	SS	Reversal exhausts; new range established	HIGH
RWL	BS (new dir)	Reversal breaks through opposing zone	MODERATE
WLS	BS	A bounding zone of the storm finally fails under pressure	VERY HIGH
WLS	SS	Volatility diminishes; oscillation range narrows	MODERATE
PBS	R/SS	Piercing move exhausts at distant Structural Extreme	HIGH
PBS	PRWL	Piercing move overextends; violent reversal	MODERATE
PRWL	SS	Reversal exhausts; new consolidation range established	HIGH

OPERATOR EMPHASIS: The WLS→BS transition is VERY HIGH confidence. This is one of the most reliable predictions in the system. When you see a Whiplash Storm, the breakout direction is determined by which bounding zone has the highest ZFP + most breakout pre-conditions active.

3.6 IP Zone Anchoring Rules (Critical Operator Teaching)

The foundational principle the operator taught: macro shapes are ANCHORED to IP zones. Without IP zone data, shape prediction is speculative. With IP zone data, shape prediction becomes DETERMINISTIC.

Shape	IP Zone Role	Engine 1 Metrics Required	Forward Mapping Logic
SS	Ceiling and Floor zones define the range	Both bounding zones: ZFP < 30%, Confluence >= 6, Regime = Consolidation/Momentum	If both boundary zones hold, SS continues. If one zone ZFP > 45%, prepare for BS transition.
BS	The failing zone triggers BS; target zone determines magnitude	Failing zone: ZFP > 45% with breakout pre-conditions. Target: Measured Move Projection anchored to Structural Extreme Magnet	If failing zone ZFP exceeds 45% with 2+ breakout signals, BS is imminent. Target is nearest Structural Extreme Magnet, not blind projection.
R	Mid-range zone provides temporary friction	Recovery zone: ZFP 20-40%, typically Post-Expiry Ghost or One-Touch. Not a Hard Wall.	Recovery is temporary. If next zone has ZFP < 30%, expect R→WL/RWL. If ZFP > 45%, expect R→BS (second leg).
WL/RWL	Hard Wall or Institutional Magnet rejects the move	Rejection zone: ZFP < 20%, Confluence >= 8, Liquidity Sweep Target from Pillar 5 Step 0	Whiplash is guaranteed at Hard Walls with ZFP < 15%. LSZ designation is the primary flag.
WLS	Cluster of 3+ zones in narrow range creates oscillation cage	Zone cluster: 3+ IP zones within 0.30-0.50 DXY, each ZFP 15-40%, NY Cut option cage amplifying	WLS continues until one bounding zone fails. Zone most likely to fail = highest ZFP + most breakout pre-conditions. WLS→BS is highest-confidence transition.
PBS	Multiple intermediate zones fail in sequence; distant Structural Extreme is target	All intermediate zones: ZFP > 50%. Target: deep Structural Extreme Magnet	PBS occurs when failing zone ZFP > 60% and Measured Move points to distant verified Structural Extreme. Each intermediate zone pierced adds momentum (cascade effect).

3.7 Macro Shape Chain — What Was Semi-Accurate and What Needs Fixing

The operator identified specific areas where the Macro Shape Chain was semi-accurate:

1. SHAPE IDENTIFICATION was mostly correct — the system correctly identified when the market was in SS, BS, R, or WL phase based on IP zone interactions. The taxonomy itself is sound.

2. CHAIN SEQUENCING had errors — the system sometimes predicted BS→R→PBS when the correct chain was BS→RWL→SS→WLS. This happened because the system did not properly account for the strength of the IP zone at the BS target. If the target zone has ZFP < 20%, the BS will be rejected (RWL), not continue (R/PBS). The chain must be conditioned on the target zone's ZFP.

3. TIMING was sometimes off — the system correctly identified which shapes would occur but sometimes misestimated when transitions would happen. The operator taught that the NY Cut Spring Release effect (10:00 AM → 10:01 AM) is the primary timing trigger for shape transitions. Transitions cluster around this time because the option cage effect breaks at 10:01 AM.

4. The Double Whiplash Pattern within SS was not being identified — the operator noted that two rapid whiplash moves within an SS range often precede a breakout attempt. This micro-pattern should be flagged as "SS (Double Whiplash)" to signal increased breakout probability.

5. The WLS→BS transition timing was not being properly predicted — the operator taught that when a Whiplash Storm's bounding zone begins showing increasing ZFP (moving from 25% toward 40%+), the breakout is approaching. The ZFP trend within WLS is the leading indicator for breakout direction and timing.

3.8 Forward Mapping Procedure (Operator-Verified)

Step 1: Current Shape Identification

Apply the Shape Classification Decision Matrix to current price position relative to IP Zone Map. This is the starting shape of the chain.

Step 2: Transition Probability Assessment

For the current shape, consult Transition Rules to identify all possible next shapes and their IP zone triggers. Evaluate whether trigger conditions are met by the IP Zone Map.

Step 3: Chain Extension

Extend the chain forward by applying transition rules iteratively. For each predicted shape, identify the IP zone trigger for the next transition. Continue until (a) end of trading week, (b) chain enters terminal SS (no zone ZFP > 35%), or (c) prediction confidence drops below MODERATE.

Step 4: Chain Confidence Scoring

Each prediction receives confidence = product of individual transition confidences adjusted for historical frequency. HIGH=0.80, MODERATE-HIGH=0.65, MODERATE=0.50, LOW-MODERATE=0.35, LOW=0.20. If any transition confidence drops below MODERATE, flag as [LOW CONFIDENCE — consider alternative chains].

Step 5: Alternative Chain Analysis

For every primary chain prediction, generate at least one alternative chain that activates if the primary fails at a specific transition point. The IP zone that determines which chain activates is the "Decision Zone."

OPERATOR EMPHASIS: Every chain extension MUST document the audit trail: current shape, IP zone evaluated as trigger, specific ZFP and breakout pre-conditions, predicted next shape, and confidence level. This audit trail ensures every prediction is independently verifiable from Engine 1 IP Zone Map data alone.

4. Actionable Items for the Next Engineer

4.1 Engine 3 — Must-Do Before Re-Enabling

1	Implement Filter Checkpoint (D1) — this is the blocking dependency. Without it, Engine 3 will consume contaminated data.
2	Implement Multi-Setup Architecture (P2) — generate ALL valid setups across ALL zones simultaneously.
3	Implement Counter-Trend Check (P1) — when price is within 0.20 DXY of a Hard Wall zone, evaluate mean-reversion probability that can override trend-following checks.
4	Implement Intra-Zone Key Level Scan (Step 0.6) — identify micro-levels within each IP zone for precise entry/SL anchoring.
5	Implement Regime-Conditional TP Logic — default TP = nearest opposing zone/LSZ edge for SS regime. Fib wormhole only as extended target if zone is breached.
6	Implement LSZ-Anchored SL (R3.3) and IP Zone-Anchored TP (R3.4).
7	Implement RRR Gate (R3.5) — minimum 2.0:1 trend, 2.5:1 counter-trend.
8	Implement Confidence Scoring (R3.9) — only setups scoring >= 65/100 are included.
9	Implement Setup Limit (R3.10) — maximum 3 setups per report.

4.2 Engine 2 — Must-Fix for Full Accuracy

1	Chain sequencing must be conditioned on target zone ZFP — if BS target has ZFP < 20%, predict RWL not R/PBS.
2	NY Cut Spring Release timing must be incorporated into shape transition timing predictions.
3	Double Whiplash Pattern within SS must be identified and flagged.
4	WLS→BS transition must use ZFP trend within WLS as leading indicator.
5	R→SS reclassification must be enforced when stabilization persists > 1 full session.
6	Alternative chain analysis must always be generated alongside primary chain.

4.3 Items Needing Review

1	Engine 3 6-phase architecture needs forward-testing on a live trading week before being made production standard.
2	Pillar 7 data (CME FX OI + ICE DXY Options OI) remains NO DATA — confluence floor should be adjusted to 5/12 when Pillar 7 is unavailable.
3	The May 18-22 case study should be used as a validation test: generate the chain prediction with and without the 99.43-99.67 zone mapped, and verify the corrected chain matches actual outcomes.
4	Post-generation compliance sweep (D6) must include a shape-chain-specific check: verify that all shape predictions reference specific IP zone triggers with ZFP values.

5. Reference Documents

Document	Path	Content
System Prompt v5.0.0	/home/z/my-project/download/Trident_Trading_Engine_System_Prompt_v5.0.0.md	Complete executable system prompt with all fixes
System Prompt v4.5.1	/home/z/my-project/download/Leviathan_Quant_System_Prompt_v4.5.1.md	Previous version (reference for changes)
Engine 3 Changelog Spec	/home/z/my-project/download/Leviathan_System_Prompt_Changelog_Engine3_Spec.pdf	Part A: Filter changes + Part B: Engine 3 rebuild spec
Engine 2 Spec	/home/z/my-project/download/Leviathan_Engine2_Macro_Shape_Chain_Map_Specification.pdf	Complete Engine 2 specification
Postmortem Report	/home/z/my-project/download/Leviathan_PostMortem_Apr20-24_2026.pdf	Full postmortem with zone grades, trade setup failure, missed trades
Derailment Report	/home/z/my-project/download/Leviathan_Derailment_Failure_Report.pdf	7 derailments from May 25-29 report
Engineer Context Handover	/home/z/my-project/download/Leviathan_Engineer_Context_Handover.pdf	Full development history and all mistakes/solutions
BULLETPROOF Report	/home/z/my-project/download/Leviathan_Consolidated_Report_BULLETPROOF.py	Working 4-pair report script
Template v2.0	/home/z/my-project/persistent_memory/Leviathan_Report_Template_v2.0.py	52 ironclad formatting rules